

Calculus in Motion: 3D VR Exploration of Solids of Revolution

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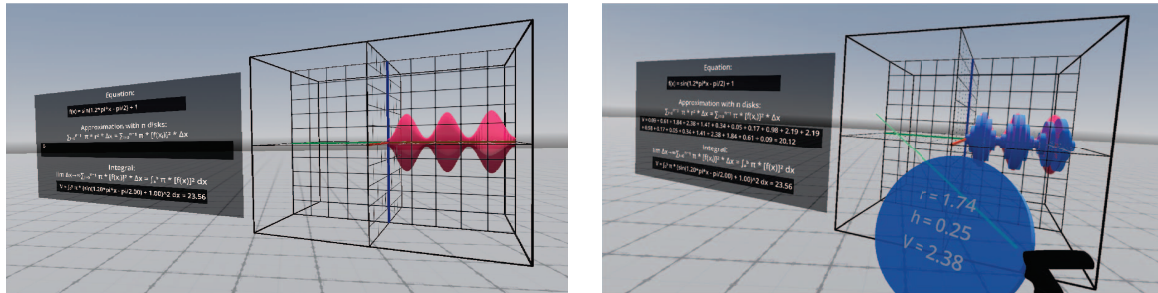


Figure 1: One feature of our VR application: the user draws a sinusoidal curve which is rotated around the x-axis to form a solids of revolution (left). The solid's volume is approximated using a finite number of disks which the user can grab to see the disk's radius, height, and volume (right). The menu on the left side of both images displays the curve equation, along with the discrete representation (disk approximation) and the continuous representation (integration) of the solid's volume.

ABSTRACT

The integration of Virtual Reality (VR) into mathematics education is an expanding area of interest driven by the importance of visualizations for understanding mathematical concepts. We hope to contribute to this area by focusing on the domain of multidimensional calculus, specifically solids of revolution. Through our VR application, we aim to deepen understanding through interactive features for 3D visualizations. By directly addressing what we have identified as solids of revolution's three most challenging-to-visualize concepts, our tool seeks to enhance users' comprehension and improve engagement in this complex mathematical domain.

Index Terms: Human-centered computing—Visualization—Visualization application domains; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Given the abstract nature of mathematical concepts, visualizations are a crucial aspect of mathematics education [6]. In pursuit of visualization solutions, many educators have turned to VR to assist in explaining mathematical concepts. VR has been shown to be an effective tool for improving learning outcomes [7]. The immersive environments of VR, coupled with its high levels of interactivity and 3D spatial visualizations, contribute to the advantages of VR for mathematics education.

We propose leveraging the benefits of VR explaining mathematics in the domain of multidimensional calculus. Visualizations are especially beneficial in the context of multidimensional calculus due to frequent engagement with variables and surfaces in 3D space.

In our VR exploration aimed at visualizing multidimensional calculus, we have chosen to concentrate on solids of revolution. This

was motivated by the importance of understanding the calculation behind the volume of these solids. These solids are essential in the engineering and manufacturing fields and have been a conventional feature of computer-aided design since the 1970s [8].

To effectively teach solids of revolution, we have identified three primary challenging-to-visualize concepts [1]:

Graphing equations: translating back-and-forth between an algebraic equation and a graph

Curve to Solid: converting a curve in 2D space into a solid in 3D space by rotating the curve around an axis

Discrete and continuous representations: approximating the volume of a solid as a discrete summation of disks as opposed to a continuous summation using integration

Unfortunately, there is limited literature on visualizing solids of revolution using AR/VR. We aim to improve on some of the existing work [5, 4] by providing more intuitive interactions in 3D space and by focusing on features for the three primary challenging concepts.

2 DESIGN AND IMPLEMENTATION

We created our application for VR headsets using Godot 4. Our application provides solutions to our identified three primary challenging concepts involved in visualizing solids of revolution. In the proposed application, the user enters a virtual space where they can freely move and interact with a 3D coordinate system presented in a cube, which will also house the generated solid. The coordinate system is placed in a cube so the user can easily and intuitively manipulate, rotate, and scale the system. For easy viewing of the underlying mathematics, a menu, which we will refer to as the equation menu, is attached to the left side of the cube. The equation menu displays the equation the user wants to graph, along with the discrete and continuous representations of the volume of the solid generated from the equation.

The user can access different functionalities of the application by using ray-casting to select icons on a toolbar located next to the user's left hand. This icon toolbar enables users to quickly switch between different functionalities. There are three icons: the first icon opens a menu for defining equations, the second icon opens a menu for adjusting settings related to the curve rotation and the discrete and continuous representations, and the third icon resets the

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scene. Observe the application's initial appearance in Figure 2.

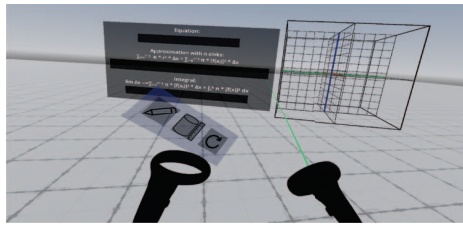


Figure 2: The application scene at start up. Displays the equation menu, coordinate system, and icon toolbar

2.1 Graphing Equations

A key feature of the application is the equation definition process. To define the equation, the user has two different input options: (1) Free-form Curve Drawing, or (2) Handwriting Recognition. Both input methods use ray-casting to draw on a virtual drawing pad. The user can switch between the two input options through a button at the top of the drawing pad menu. See Figure 3.

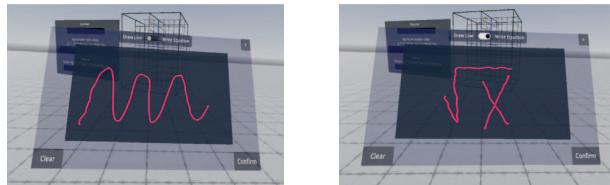


Figure 3: Two ways to graph equations: Free-form Curve Drawing (left) and Handwriting Recognition (right)

2.1.1 Free-form Curve Drawing

When the user has a specific curve in mind but is uncertain of the curve's equation, we allow the user to draw the curve on the drawing pad. This feature supports lines, parabolic curves, and sinusoidal curves. To determine the equation of the drawn curve, the Godot script executes a command, which runs a Python script that leverages libraries to perform accurate curve-fitting. The resulting equation is transmitted back to Godot by redirecting the Python standard output.

Importantly, the equation is then displayed on the equation menu. This ensures that the user can directly view the mathematical equation that matches the shape of their drawn curve to gain a tangible understanding of the relationship between a curve's visual form and its underlying mathematical representation.

2.1.2 Handwriting Recognition

We also wanted to address when the user has a specific equation in mind. To avoid having users type their equations using an unwieldy keyboard, we allow the user to hand-write their equation. We once again setup communication between Godot and Python. Adapting code from Sabari Nathan [3], we used optical character recognition (OCR) to convert the user's equation into individual symbols. We also trained our own model by adapting Nathan's Convolutional Neural Network (CNN) on a handwritten math symbols dataset of over 100,000 image samples [2]. Using this model, we predict the equation based on the user's handwritten equation. The predicted equation is then displayed on the equation menu.

2.2 Curve to Solid

After the equation is defined, it is used to create the solid of revolution. The equation defined curve is rotated around an axis resulting in a unique 3D solid. To help users visualize the transformation of the curve into a solid, we have incorporated an animation feature that rotates the user's curve around the axis in real-time, allowing the user to follow the solid generation process. See Figure 4. After

the animation, the user can also manually rotate the curve around the axis for a hands-on approach for understanding solid generation.

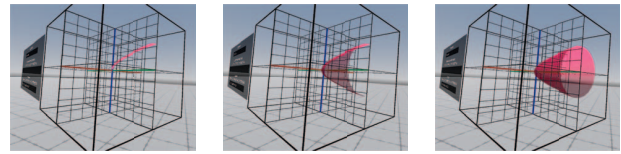


Figure 4: Animation of the rotation of the curve $f(x) = \sqrt{x}$

2.3 Discrete and Continuous Representations

Once the solid has been generated, the user can adjust settings related to the discrete and continuous representations. The discrete representation consists of a finite number of disks that, when added together, approximate the continuous representation—the solid's volume. We overlay the disks on top of the generated solid and allow the user to grab the disks. Each face of a disk has the disk's radius, height, and volume. Additionally, on the equation menu, we show the sum of the volumes of each disk to approximate a final volume. We also display the integration or the mathematical expression of the continuous representation which can be compared to the approximated volume. The user is able to adjust the number of disks to observe as the number of disks increase, the approximation of the volume becomes more accurate both visually within the cube and mathematically on the equation menu. See Figure 1.

3 CONCLUSION

In this paper, we propose a novel VR application for interactive 3D visualizations to explain the mathematical concept, solids of revolution. We distinguish ourselves from previous attempts by implementing user-friendly features that directly address the three primary challenging concepts. These features include enabling curve drawing and handwritten equations to grasp the relationship between curve shape and equation, animating and providing a manual method for curve rotation to illustrate solid generation, and providing visual and mathematical demonstrations of discrete and continuous representations of the solid's volume. We hope to inspire more VR applications for visualizing and understanding other complex mathematical concepts.

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